

Supplemental figure legends

Fig.S1: IDO1 enzymatic activity is induced by NT and GD2.CAR T-produced IFN γ . **A** and **B**) Concentration of kynurenines (Kyns) in the supernatant, measured by HPLC at 24 and 48 Hours. Data are expressed as mean $\pm$ SEM. **C**) FACS analysis of GD2 expression in ACN and IMR-32 cell line. **D**) WB analysis of IDO1 (exogenous) expression in ACN and IMR-32 IDO1 overexpressing clones, compared with Ctrl conditions transfected with empty vector. HPLC analysis of co-culture supernatants between NT cells and ACN cell line (**E**), and IMR-32 cell line (**G**); between GD2.CAR T cells and ACN cell line (**F**) and IMR-32 cell line (**H**). Target cells were tested as Ctrl condition (Ctrl), Ctrl with IDO1 inhibitor (Ctrl-BMS), IDO1-overexpressing (IDO1) and IDO1-overexpressing cells with IDO1 inhibitor (IDO1-BMS). Time of co-culture analyzed: 3, 5 and 7 days

Fig.S2: NK cells isolated by human healthy donor peripheral blood are activated against K562 cell line. **A**) Representative plot of Annexin V/PI evaluation of K562 cells in co-culture with NK cells at 24 hours (left) and at day 3. K562 cells were marked in Red by PKH26 cell linker to be identified in the co-culture (**B**) Graph of % Annexin V-positive cells mean values of the experiments shows in A